

Stories We Tell (and the Machines Retell)

How a Large Language Model changes folktales
when retelling them across cultures

Areg Karapetian

Università degli Studi di Milano
Natural Language Processing — Prof. Alfio Ferrara
NLP Final Project P12

July 2026

What is this project about?

- We take **30 folktales** from three cultural traditions: European, Asian, and African Diaspora.
- We ask a language model (**Llama 3.1 8B**) to retell each story *“in the style of”* the other two cultures.
- Then we **measure what the model changed**: only the words — or the story structure itself?

The question: words or structure?

When the model retells a folktale in another cultural style, it can do two different things:

Surface change

New names, new places,
new vocabulary. . .

same plot skeleton

Deep change

Characters merged or dropped,
events reorganised. . .

the skeleton itself changes

Reading the texts cannot tell these apart — the words change in both cases.

So we need to measure structure separately from words.

The data

- **30 folktales:** 10 European, 10 Asian, 10 African Diaspora
- From public-domain folklore collections
- All in English, 320–900 words each

From 30 stories to 90 versions

Each original is retold into the **two other** styles:

$30 \times 2 = 60$ retellings $\Rightarrow 30 + 60 = \mathbf{90}$ story versions

One language, one model $\Rightarrow$ between an original and its retelling,
the target culture is the only thing that changes.

Step 1 — Generate the retellings

- Model: **Llama 3.1 8B**, running locally via **Ollama**
- **Zero-shot prompt** — no examples, because we want to measure what the model *itself* thinks each cultural style is

The key instruction

*“Keep the SAME main events and the SAME ending.
Do NOT add new major plot points.”*

- We explicitly **forbid** changing the plot — so every structural change we find later, the model made **despite the instruction**
- Length kept within $\pm 15\%$ of the original (a summary is not a retelling)
- Temperature 0.35 — some stylistic freedom, few hallucinations

Step 2 — Extract the plot events

Every one of the 90 versions is reduced to **10–14 plot events**
(short sentences, chronological order, strict JSON):

Example

1. The king orders his three sons to seek their fortunes.
2. Don Juan meets an old man on the road.
3. Don Juan marries Chonguita. ...

Why this step matters:

- removes all stylistic decoration — **only the plot remains**
- same number of events for every story $\Rightarrow$ fair comparison
- low temperature (0.2) + retries + fallback parsing
 $\Rightarrow$ **all 90/90 versions extracted successfully**

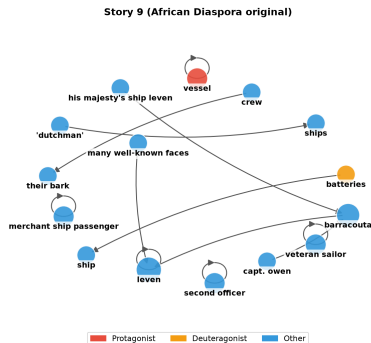
Step 3 — From events to a story skeleton

Each event becomes a **triple**:
(*subject, relation, object*)

“The king ordered his sons”
⇒ (the king, ordered, his sons)

The triples form the **narrative graph**
— the story skeleton:

- characters and objects = **nodes**
- events = **directed, labelled edges**
(who does what to whom)
- roles from pure structure:
most active subject = **Protagonist**,
second = **Deuteragonist**, rest = Other



A real narrative graph from our data
(self-loops = events with no object)

Step 4 — Measure the difference: four metrics

For each of the 60 pairs (original vs. its retelling):

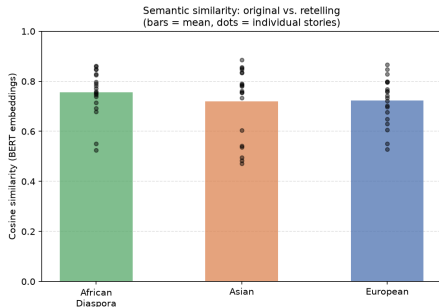
1. **Graph edit distance (GED)** — how many changes turn one skeleton into the other? *shape*
2. **Relation overlap** — do both versions use the same action words? *actions*
3. **Role alignment** — is the protagonist still acting at the same points of the plot? *characters*
4. **Semantic similarity** — cosine similarity of BERT embeddings of the full texts *meaning*

Results — the model treats cultures differently

Target culture	GED ↓	Rel. overlap ↑	Subj. align. ↑	Obj. align. ↑
African Diaspora	0.228	0.136	0.460	0.708
Asian	0.269	0.115	0.500	0.500
European	0.318	0.121	0.461	0.512

- **European** retellings restructure the story the most (highest GED)
— even though the prompt forbade it
- **African Diaspora** retellings stay closest on every metric
- **Subject-role alignment is the same everywhere** (≈ 0.5):
the model never loses track of *who the story is about*

Results — meaning is preserved, and confirms the picture



- Independent check with **BERT embeddings**
(no graphs involved)
- Same ordering: African Diaspora closest (0.756),
European and Asian ≈ 0.72
- Two independent method families agree
 $\Rightarrow$ the asymmetry is real, not an artifact

Thank you!